

Notes: Caldara and Iacoviello (AER, 2022)

Measuring Geopolitical Risk

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1 Big picture

- **Question.** How can researchers measure geopolitical risk consistently over time, and what are the real effects of such risk on macroeconomic activity, tail outcomes, and firm investment?
- **Definition.** Geopolitical risk is defined as the threat, realization, and escalation of adverse events associated with wars, terrorism, and tensions among states or political actors that affect the peaceful course of international relations.
- **Core measurement idea.** The paper builds a news-based geopolitical risk index by counting the share of newspaper articles that discuss adverse geopolitical events and associated risks.
- **Main index.** The recent GPR index starts in 1985 and uses ten leading newspapers from the United States, United Kingdom, and Canada. The historical GPR index starts in 1900 and uses the Chicago Tribune, the New York Times, and the Washington Post.
- **Main spikes.** The index spikes around World War I, World War II, Pearl Harbor, D-Day, the Korean War, the Cuban Missile Crisis, the Gulf War, 9/11, the Iraq War, the Paris terrorist attacks, and the US-North Korea and US-Iran tensions.
- **Threats versus acts.** The authors split the index into geopolitical threats (GPT) and geopolitical acts (GPA). Threats capture risks, warnings, military buildups, nuclear threats, and terrorist threats. Acts capture the beginning or escalation of war and terrorist attacks.
- **Validation.** The automated index is validated against a manually constructed narrative index based on 44,000 New York Times front pages, country-specific indices, military spending news, war deaths, financial volatility, and human audits of newspaper articles.
- **Macroeconomic result.** A two-standard-deviation shock to the GPR index predicts lower US investment, hours, stock prices, and oil prices. Fixed investment falls by about 1.5 percent after roughly one year, and hours fall by about 0.6 percent.
- **Threats matter.** Both realized geopolitical acts and unrealized geopolitical threats generate contractionary effects. The threat channel is important because economic agents may reduce investment and hiring even before adverse events fully materialize.
- **Tail-risk result.** In a 26-country panel from 1900 to 2019, higher geopolitical risk is associated with a higher probability of economic disaster and larger downside risks to future GDP growth.
- **Firm-level result.** Firms in industries more exposed to aggregate GPR reduce investment more after GPR shocks. Firms that discuss more idiosyncratic geopolitical risk in earnings calls also cut investment.
- **Main contribution.** The paper contributes a long-run, transparent, replicable measure of geopolitical risk and shows that geopolitics is not just a political or financial-market concern; it is systematically connected to investment, employment, disaster probability, and firm decisions.
- **Takeaway.** Geopolitical risk behaves like a macro-financial shock that affects both average outcomes and downside risks. Its real effects come from both actual events and the fear of future events.

2 Data and measurement (key objects)

2.1 Why this setting is useful

- Geopolitical risk is hard to measure because it involves expectations, threats, realized events, escalation, and media perception.
- Newspapers provide a high-frequency record of contemporaneous perceptions: the index measures what the press, public, investors, and policymakers were likely seeing in real time.
- The long historical sample makes it possible to study rare but important episodes, including wars and economic disasters.
- The paper also creates country-, industry-, and firm-level versions of the measure, which allows it to move from aggregate time-series evidence to cross-sectional exposure evidence.

Interpretation. The index is not just a count of wars. It is a measure of how much adverse geopolitical risk was salient in the news at a given point in time.

2.2 Definition of geopolitical risk

The paper defines geopolitical risk as:

The threat, realization, and escalation of adverse events associated with wars, terrorism, and any tensions among states and political actors that affect the peaceful course of international relations.

This definition has three important parts:

- **Threat:** risk of war, terrorism, military confrontation, nuclear conflict, or disruption of peace.
- **Realization:** actual outbreak of war, terrorist attack, military strike, invasion, or similar event.
- **Escalation:** worsening of an existing conflict, military buildup, sanctions, blockades, or intensification of tensions.

Interpretation. The word “risk” is used broadly. The index captures both ex ante fear and ex post realization. This is why the paper separately constructs threat and act components.

2.3 Newspaper data

The authors use roughly 25 million newspaper articles from 1900 onward.

- **Recent index, 1985 onward:** automated searches in ten newspapers: Chicago Tribune, Daily Telegraph, Financial Times, Globe and Mail, Guardian, Los Angeles Times, New York Times, USA Today, Wall Street Journal, and Washington Post.
- **Historical index, 1900 onward:** searches in the historical archives of the Chicago Tribune, New York Times, and Washington Post.

- **Frequency:** the main indices are monthly; the authors also construct a daily index.
- **Normalization:** the index is normalized to have mean 100 over the relevant baseline period.

Interpretation. The recent index is broader in newspaper coverage, while the historical index is longer. Their overlap is highly correlated, which supports the use of the historical series.

2.4 Headline GPR index

Let A_t be the set of newspaper articles in month t . Let $\mathbf{1}\{a \in \mathcal{G}\}$ equal one if article a satisfies the geopolitical-risk search query. A simplified representation of the index is:

$$\text{GPR}_t = 100 \times \frac{\frac{1}{|A_t|} \sum_{a \in A_t} \mathbf{1}\{a \in \mathcal{G}\}}{\bar{s}}, \quad (1)$$

where $\bar{s}$ is the sample-average article share used for normalization.

Interpretation. The raw object is the fraction of newspaper articles discussing adverse geopolitical events. The normalization makes values easier to compare over time.

2.5 Search categories

The search query is organized into eight categories.

- **Threat categories:**

1. war threats,
2. peace threats,
3. military buildup,
4. nuclear threats,
5. terrorist threats.

- **Act categories:**

6. beginning of war,
7. escalation of war,
8. terrorist acts.

The dictionary combines topic words such as war, military, nuclear, terrorism, and actors with threat or act words such as threat, fear, risk, crisis, buildup, attack, invasion, strike, and kill. The authors also exclude words that create false positives, such as movie, film, book, memorial, sport, and real estate.

Interpretation. The query is intentionally stricter than a one-word count. It usually requires the joint occurrence of geopolitical topic words and threat or act words.

2.6 Threats and acts components

The headline index can be decomposed as:

$$\text{GPR}_t \approx \text{GPT}_t + \text{GPA}_t, \quad (2)$$

where:

- GPT_t is the geopolitical threats index, based on categories 1 through 5;
- GPA_t is the geopolitical acts index, based on categories 6 through 8.

Interpretation. The decomposition matters because threats and realized acts can have different economic mechanisms. Threats mainly affect expectations and precautionary behavior; acts may also destroy capital, disrupt production, or change government spending.

2.7 Country-specific GPR

For a country i , the country-specific index counts articles that both satisfy the GPR query and mention the country, its capital, or a major city:

$$_{i,t} = \frac{\#\{\text{GPR articles mentioning country } i \text{ in } t\}}{\#\{\text{all articles in } t\}}. \quad (3)$$

Interpretation. Country-specific GPR captures the US newspaper perspective on risks posed by or involving that country. It allows the paper to exploit regional events that are too small to dominate the global index.

2.8 Validation objects

The paper validates the index using several external and internal checks:

- a manually coded narrative GPR index based on New York Times front pages;
- a human audit of a sample of newspaper articles;
- comparison with historical episodes and the largest index spikes;
- country-specific versions that line up with country-specific geopolitical events;
- comparison with Ramey’s military spending news;
- comparison with worldwide deaths from war and terrorism;
- comparison with the VIX and the Economic Policy Uncertainty index;
- robustness to different search rules, newspaper choices, and exclusion words.

Interpretation. Validation is central because the paper is first a measurement paper. The empirical results are only useful if the index is meaningful.

2.9 Macroeconomic data

The US VAR uses quarterly data from 1986:I through 2019:IV. The main variables are:

- log GPR index,
- VIX,
- log real business fixed investment per capita,
- log private hours per capita,
- log real S&P 500 index,
- log real WTI oil price,
- two-year US Treasury yield,
- Chicago Fed National Financial Conditions Index.

Interpretation. The VAR asks whether innovations to GPR predict the joint movement of financial uncertainty, asset prices, investment, and labor input.

2.10 Cross-country disaster and growth data

The tail-risk analysis uses annual data for 26 countries from 1900 through 2019.

- Economic disasters are based on updated Barro-Ursua and Nakamura et al. disaster data.
- The authors study GDP growth, TFP growth, and military expenditures.
- The panel combines global GPR and country-specific GPR.

Interpretation. The long panel allows the paper to study rare outcomes, not just ordinary business-cycle fluctuations.

2.11 Firm and industry data

The firm-level analysis uses three layers of data:

- **Industry exposure:** daily returns of 49 Fama-French industry portfolios regressed on daily changes in GPR.
- **Firm investment:** Compustat capital expenditures scaled by previous-period property, plant, and equipment.
- **Firm-level GPR:** text analysis of earnings-call transcripts from 2005:I through 2019:IV, counting risk words near geopolitical words.

Interpretation. The industry analysis uses market prices to infer which industries are exposed to geopolitical risk. The firm analysis uses managers' own earnings-call language to detect idiosyncratic geopolitical concerns.

3 Models and empirical specifications (all equations + interpretation)

3.1 Index-construction model

The basic object is the article share:

$$s_t = \frac{1}{N_t} \sum_{a=1}^{N_t} \mathbf{1}\{\text{article } a \text{ matches GPR query}\}. \quad (4)$$

The normalized index is:

$$\text{GPR}_t = 100 \times \frac{s_t}{\bar{s}}. \quad (5)$$

Interpretation. A value above 100 means that newspaper attention to geopolitical risk is above its baseline average. A spike means geopolitical risk is unusually salient in contemporaneous news coverage.

3.2 Aggregate VAR specification

The benchmark VAR is:

$$Y_t = c + A_1 Y_{t-1} + A_2 Y_{t-2} + u_t, \quad (6)$$

where Y_t contains the eight variables listed above. The GPR shock is identified with a Cholesky decomposition of the reduced-form residual covariance matrix, ordering GPR first.

Interpretation. Ordering GPR first means that contemporaneous correlations between GPR and macro-financial variables are attributed to GPR moving the other variables, not to the other variables moving GPR within the quarter.

3.3 Acts-versus-threats VAR

The acts-versus-threats VAR replaces the headline GPR index with GPA_t and GPT_t :

$$Y_t^{AT} = (\text{GPA}_t, \text{GPT}_t, \text{investment}_t, \text{hours}_t, \dots). \quad (7)$$

The paper orders GPA first and GPT second. This allows an acts shock to move threats contemporaneously, while a threats shock captures threats that do not immediately materialize into acts.

Interpretation. This design separates two channels: actual adverse events and fear of future adverse events. The counterfactual responses hold one component constant to ask what each channel contributes in isolation.

3.4 Disaster-probability regression

The disaster specification is:

$$D_{i,t} = \alpha_i + \beta \text{GPR}_t + \gamma_{i,t} + \delta \Delta \text{GDP}_{i,t-1} + \text{controls}_{i,t} + u_{i,t}, \quad (8)$$

where $D_{i,t}$ is a disaster dummy for country i in year t .

Interpretation. The coefficient β measures the association between global geopolitical risk and disaster probability. The coefficient γ measures whether country-specific geopolitical risk is linked to disaster probability in that country.

3.5 Quantile-growth regressions

The quantile model is:

$$\mathcal{Q}_\tau (\Delta y_{i,t+1} \mid x_{i,t}) = \alpha_\tau + \beta_{\tau i,t}, \quad (9)$$

where $\Delta y_{i,t+1}$ is one-year-ahead GDP growth, TFP growth, or military expenditure.

Interpretation. Estimating β_τ at the tenth, fiftieth, and ninetieth quantiles asks whether geopolitical risk changes the whole distribution of future outcomes, not just the mean.

3.6 Firm-level decomposition of geopolitical risk

The paper conceptualizes firm-level geopolitical risk as:

$$\text{GPR}_{i,t} = \text{GPR}_t + \text{GPR}_t \Lambda_k + Z_{i,t}, \quad (10)$$

where i indexes firms and k indexes industries.

- GPR_t : aggregate geopolitical risk.
- $\text{GPR}_t \Lambda_k$: industry-specific exposure to aggregate geopolitical risk.
- $Z_{i,t}$: idiosyncratic firm-level geopolitical risk.

Interpretation. This decomposition says that a firm can be affected by geopolitical risk because the whole economy is affected, because its industry is especially exposed, or because the firm has its own geopolitical exposure.

3.7 Industry exposure from stock returns

Industry exposure is estimated using daily data:

$$R_{k,t} = \alpha_k + \beta_k \Delta \text{GPR}_t + \varepsilon_{k,t}, \quad (11)$$

where $R_{k,t}$ is the daily excess return of industry k . The estimated β_k coefficients are demeaned and multiplied by minus one so that larger values mean greater exposure.

Interpretation. If an industry's stock returns fall more when GPR rises, that industry is classified as more exposed to geopolitical risk.

3.8 Industry-exposure local projection

The dynamic response of firm investment to aggregate GPR through industry exposure is estimated with:

$$\log ik_{i,t+h} = \alpha_{i,h} + \beta_h (\mathbb{D}_k \Delta \log \text{GPR}_t) + d_h X_{i,t} + \varepsilon_{i,t+h}, \quad (12)$$

where ik is investment, $\mathbb{D}_k$ is the high-exposure industry dummy, and $X_{i,t}$ includes cash flow, Tobin's Q , and lagged investment.

Interpretation. The sequence β_h traces how much more investment falls for exposed industries after an aggregate GPR shock.

3.9 Firm-specific local projection

The idiosyncratic firm-risk specification is:

$$\log ik_{i,t+h} = \alpha_{i,h} + \alpha_{k,t,h} + \gamma_h Z_{i,t} + d_h X_{i,t} + \varepsilon_{i,t+h}. \quad (13)$$

The sector-by-quarter fixed effect $\alpha_{k,t,h}$ absorbs aggregate and industry-specific shocks.

Interpretation. The coefficient γ_h captures whether a firm that mentions more geopolitical risk in its earnings call reduces investment relative to other firms in the same sector and quarter.

4 Identification and instruments (what makes the estimates credible)

4.1 No single conventional instrument

The paper does not rely on one external instrument for geopolitical risk. Instead, credibility comes from:

- careful construction of a transparent news-based index;
- validation against independent historical and narrative evidence;
- time-series identification in the VAR;
- cross-country variation in country-specific GPR;
- cross-industry exposure based on high-frequency stock-market reactions;

- firm fixed effects and sector-by-quarter fixed effects in firm regressions.

Interpretation. This is a measurement-and-triangulation paper. The evidence is strongest because the authors show similar patterns in aggregate, country, industry, and firm data.

4.2 Measurement validity

The key threat is that the index might measure media attention rather than true geopolitical risk. The paper addresses this with several checks:

- the largest historical jumps line up with major geopolitical events;
- the automated index correlates strongly with a manually coded narrative index;
- country-specific indices spike around country-relevant events;
- the index correlates with war deaths and military spending news;
- the index is not mechanically driven by economic-policy or financial-uncertainty news;
- human audits show that the search query has meaningful accuracy.

Interpretation. These exercises make it more plausible that the index captures real geopolitical risk, not just noise in newspapers.

4.3 VAR identification

The VAR orders GPR first. This assumes that, within the quarter, geopolitical shocks can affect investment, hours, stock prices, oil prices, and financial conditions, but those variables do not contemporaneously cause GPR.

Why plausible:

- many GPR spikes are caused by political and military events that are plausibly outside the normal US business cycle;
- the dictionary is designed around geopolitical language, not economic language;
- the authors report robustness to alternative VAR designs and comparisons with financial and policy uncertainty.

Limit. Some geopolitical events may be partly endogenous to economic conditions. The VAR therefore supports predictive and shock-like interpretation, but it is not a randomized experiment.

4.4 Threats versus acts

The acts-versus-threats design is credible because it uses different search categories:

- acts include realized events such as invasions, attacks, and escalation;
- threats include warnings, risks, crises, military buildups, and nuclear threats.

Interpretation. If both components generate contractionary effects, the paper can argue that economic damage is not only caused by physical destruction or realized conflict. Fear itself can be contractionary.

4.5 Cross-country disaster evidence

The disaster regressions include country fixed effects, lagged GDP growth, global GPR, country-specific GPR, and robustness controls such as world-war dummies and period dummies.

Credibility:

- country fixed effects absorb time-invariant differences across countries;
- country-specific GPR exploits variation in which countries are involved in geopolitical events;
- robustness to world-war dummies shows that the result is not only driven by the two world wars.

Limit. Wars and disasters are often jointly determined. The regression documents a strong association between GPR and disaster risk, but it cannot fully isolate all political and economic channels.

4.6 Industry-exposure evidence

The industry design uses daily stock returns to measure exposure. Industries whose returns fall more when GPR rises are treated as more exposed.

Credibility:

- daily data reduce confounding from slow-moving industry trends;
- exposure is estimated before being used in investment regressions;
- the approach asks whether the industries that markets perceive as more exposed also cut real investment more.

Limit. The exposure measure is not a perfect instrument. Stock returns may reflect many channels, including expected demand, supply disruption, risk premia, and government spending.

4.7 Firm-level evidence

The firm-specific analysis includes firm fixed effects and sector-by-quarter fixed effects. This absorbs stable firm differences and common shocks to firms in the same sector and quarter.

Credibility:

- firm-level GPR is based on earnings-call language, capturing risks salient to managers and investors;
- sector-by-quarter fixed effects remove aggregate and industry-level geopolitical components;
- controls include cash flow, Tobin's Q, and lagged investment.

Limit. Earnings-call mentions may partly reflect managers' interpretation of bad news rather than an exogenous shock. The evidence is best read as firm-level association plus supporting mechanism evidence.

5 Tables and figures

5.1 Table 1: Search query for the GPR index

Read: The table lists the eight categories used to build the index, the topic and threat/act words, the peak historical month for each category, and the excluded words.

Economic message: The index is not a vague sentiment measure. It is a structured dictionary that separates threats from acts and reduces false positives.

TABLE 1—SEARCH QUERY FOR THE GPR INDEX					
Category	Search query	Peak (month)	Contribution to index percent		
			Full sample	1900–1959	1960–2019
Panel A. Search categories and search queries					
Threats					
1. War threats	War_words N/2 Threat_words	Germany invades Czech. (September 1938)	13.5	17.9	9.2
2. Peace threats	Peace_words N/2 Peace_disruption_words	Iran crisis of 1946 (April 1946)	3.5	4.3	2.7
3. Military buildup	Military_words AND buildup_words	Cuban Missile Crisis (October 1962)	23.5	21.3	25.8
4. Nuclear threats	Nuclear_bigrams AND Threat_words	Nuclear ban negotiations (August 1963)	10.1	4.2	16.0
5. Terrorist threats	Terrorism_words N/2 Threat_words	9/11 (October 2001)	2.7	0.3	5.0
Acts					
6. Beginning of war	War_words N/2 War_begin_words	WWII begins (September 1939)	18.8	26.8	10.7
7. Escalation of war	Actors_words N/2 Actors_fight_words	D-Day (June 1944)	19.6	23.9	15.3
8. Terrorist acts	Terrorism_words N/2 Terrorism_act_words	9/11 (September 2001)	8.3	1.3	15.2
Panel B. Search words					
Topic sets					
War_words	war OR conflict OR hostilities OR revolution* OR insurrection OR uprising OR revolt OR coup OR geopolitical				
Peace_words	peace OR truce OR armistice OR treaty OR parity				
Military_words	military OR troops OR missile* OR "arms" OR weapon* OR bomb* OR warhead*				
Nuclear_bigrams	"nuclear war*" OR "atomic war*" OR "nuclear missile*" OR "nuclear bomb*" OR "atomic bomb*" OR "h-bomb*" OR "hydrogen bomb*" OR "nuclear test" OR "nuclear weapon"				
Terrorism_words	terror* OR guerrilla* OR hostage*				
Actor_words	allie* OR enem* OR insurgen* OR foe* OR army OR navy OR aerial OR troops OR rebels				
Phrases					
Threat/act sets	Phrases				
Threat_words	threat* OR warn* OR fear* OR risk* OR concern* OR danger* OR doubt* OR crisis OR trouble* OR disput* OR tension* OR imminen* OR inevitable OR footing OR menace* OR brink OR scare OR peril*				
Peace_disruption_words	threat* OR menace* OR reject* OR peril* OR boycott* OR disrupt*				
Buildup_words	buildup* OR build-up* OR sanction* OR blockad* OR embargo OR quarantine OR ultimatum OR mobiliz*				
War_begin_words	begin* OR start* OR declar* OR begun OR began OR outbreak OR "broke out" OR breakout OR proclamation OR launch*				
Actor_fight_words	advance* OR attack* OR strike* OR drive* OR shell* OR offensive OR invasion OR invad* OR clash* OR raid* OR launch*				
Terrorism_act_words	attack OR act OR bomb* OR kill* OR strike* OR hijack*				
Panel C. Excluded words					
Exclusion words	movie* OR film* OR museum* OR anniversar* OR obituar* OR memorial* OR arts OR book OR books OR memoir* OR "price war" OR game OR story OR history OR veteran* OR tribute* OR sport OR music OR racing OR cancer OR "real estate" OR mafia OR trial OR tax				

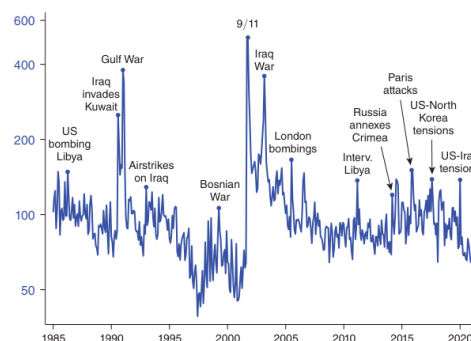


FIGURE 1. RECENT GPR INDEX FROM 1985

Notes: Recent GPR index from 1985 through 2020. Index is normalized to 100 throughout the 1985–2019 period.

5.2 Figure 1: Recent GPR index from 1985

Read: The largest recent spikes occur around the Gulf War, 9/11, and the Iraq War. Smaller spikes occur around Libya, London bombings, Crimea, Paris attacks, and US-North Korea and US-Iran tensions.

Economic message: The recent index captures events that would naturally be viewed as major global geopolitical shocks.

5.3 Figure 3: Historical GPR index from 1900

Read: The index is extremely high around World War I and World War II, with notable spikes during Pearl Harbor, D-Day, the Korean War, the Cuban Missile Crisis, the Gulf War, 9/11, and the Iraq War.

Economic message: The index passes a basic historical plausibility test: it assigns high geopolitical risk to periods that historians and contemporaries would identify as dangerous.

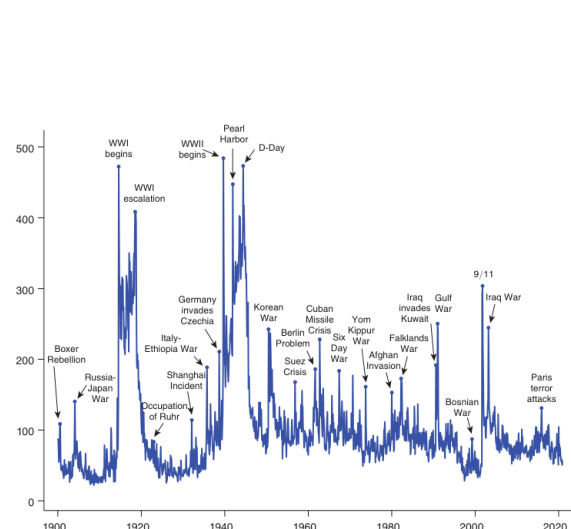


FIGURE 3. HISTORICAL GPR INDEX FROM 1900

Notes: Historical GPR Index from January 1900 through December 2020. Index is normalized to 100 throughout the 1900–2019 period.

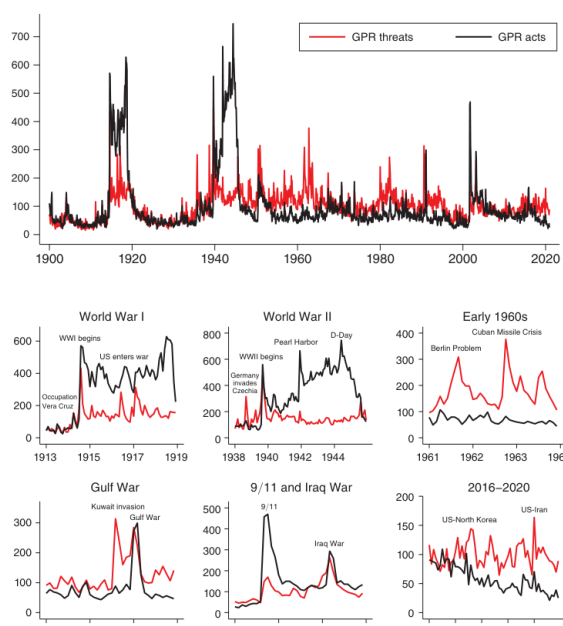


FIGURE 4. GEOPOLITICAL THREATS AND GEOPOLITICAL ACTS

Notes: Geopolitical threats (GPT) and the geopolitical acts (GPA) indices. The GPT index is constructed by searching articles in categories 1 to 5 in Table 1. The GPA index is constructed by searching articles in categories 6 to 8 in Table 1. Both indices are normalized to 100 in the 1900–2019 period.

5.4 Figure 4: Geopolitical threats and geopolitical acts

Read: Some events are dominated by realized acts, such as the outbreak of wars and 9/11. Others are dominated by threats, such as the Cuban Missile Crisis and some recent US-North Korea or US-Iran tensions.

Economic message: Threats and acts are related but not identical. The decomposition is useful for separating physical-event channels from expectations and precautionary channels.

5.5 Figure 9: Aggregate VAR responses to increased geopolitical risk

Read: A two-standard-deviation GPR shock raises the GPR index persistently. Investment falls, hours fall, stock prices fall, oil prices fall, and the VIX rises briefly.

Economic message: Geopolitical risk is associated with a broad contraction in real activity and financial markets, not only with higher uncertainty.

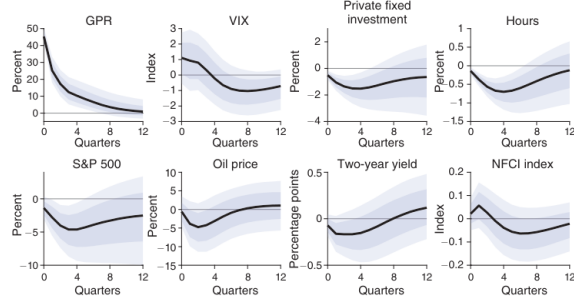


FIGURE 9. THE IMPACT OF INCREASED GEOPOLITICAL RISK

Notes: The black solid line depicts the median impulse response of the specified variable to a two standard deviation increase in the GPR index. The dark and light shaded bands represent the 68 percent and 90 percent pointwise credible sets, respectively.

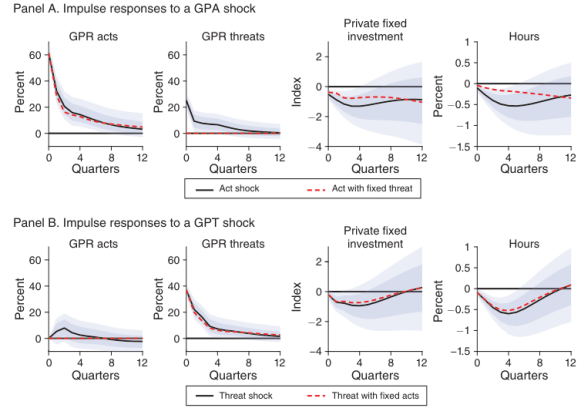


FIGURE 10. THE IMPACT OF INCREASED GEOPOLITICAL RISK: ACTS VERSUS THREATS

Notes: The black line depicts the median impulse response of the specified variable to a two standard deviations exogenous increase in the GPA index (panel A) and in the GPT index (panel B). The red dashed line depicts the outcome of a counterfactual simulation that keeps GPT (panel A) and GPA (panel B) constant. The dark and light shaded bands represent the 68 percent and 90 percent pointwise credible sets, respectively.

5.6 Figure 10: Acts versus threats in the VAR

Read: Both GPA shocks and GPT shocks reduce investment and hours. Counterfactual simulations show that each component has contractionary content even when the other component is held fixed.

Economic message: Fear and anticipation matter. The macroeconomic effects of geopolitical risk are not limited to realized destruction or direct conflict.

5.7 Table 3: Geopolitical risk and economic disasters

Read: Higher global and country-specific GPR are associated with higher disaster probability. The association remains when the paper adds country fixed effects, world-war controls, GPR-spike variables, and period dummies. High country GPR is especially related to disaster onset.

Economic message: Geopolitical risk is linked to rare but severe macroeconomic downturns, not only to ordinary cyclical fluctuations.

5.8 Table 4: Quantile regression effects

Read: Country-specific GPR reduces average future GDP and TFP growth. The lower tail is much more affected: the tenth-quantile GDP effect is about -1.44 , compared with an OLS effect of about -0.35 . Military spending rises especially in the upper tail.

Economic message: GPR increases downside macroeconomic risk. It is especially informative about bad outcomes in the left tail of growth.

5.9 Figure 11: Firm-level investment responses

Read: Firms in exposed industries reduce investment more after aggregate GPR shocks. Firms with higher idiosyncratic firm-level GPR also reduce investment, with effects accumulating over several quarters.

TABLE 3—GEOPOLITICAL RISK AND ECONOMIC DISASTERS

	Disaster (1)	Disaster (2)	Disaster (3)	Disaster (4)	Disaster (5)	Onset (6)	Ending (7)
GDP growth $t-1$	−0.0071 (0.0030)	−0.0062 (0.0030)	−0.0056 (0.0030)	−0.0065 (0.0032)	−0.0056 (0.0026)	−0.0009 (0.0010)	0.0012 (0.0010)
GPR	0.1753 (0.0223)	0.1144 (0.0241)	0.0337 (0.0469)		0.1001 (0.0236)	0.0180 (0.0237)	−0.0175 (0.0094)
Country GPR		0.0940 (0.0160)	0.0842 (0.0170)		0.0794 (0.0175)	0.0664 (0.0295)	−0.0090 (0.0105)
Dummy WWI/WWII			0.3328 (0.1761)				
GPR spikes				0.1692 (0.0246)			
Country GPR spikes				0.0821 (0.0122)			
Dummy pre-1946					0.2437 (0.0490)		
Dummy 1946–1972					0.1152 (0.0467)		
Constant	0.2309 (0.0252)	0.2289 (0.0273)	0.1947 (0.0341)	0.1762 (0.0302)	0.1112 (0.0320)	0.0401 (0.0185)	0.1180 (0.0130)
Observations	3,056	3,056	3,056	3,056	3,056	2,447	609
R^2	0.20	0.20	0.21	0.18	0.26	0.13	0.02
Countries	26	26	26	26	26	26	26
Country fixed effects	No	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Standard errors in parentheses are clustered by country and year. The table shows the effects of global and country-specific geopolitical risk on the probability of economic disaster in a panel of countries from 1900 through 2019. GDP growth is expressed in percent. GPR is standardized. Country GPR is standardized by country. Country GPR is the number of GPR articles mentioning the country divided by total number of newspaper articles. The GPR spikes variable equals GPR in the ten observations with the highest value of the GPR relative to a 20-year lagged moving average, and zero otherwise. The country GPR spikes variable equals country GPR when country GPR is larger than two standard deviations relative to a 20-year lagged moving average, and zero otherwise. The war dummy equals one in the years 1914–1918 and 1939–1945. See online Appendix D for the list of countries. Disaster episode data were constructed using updated GDP and consumption per capita data from Barro and Ursua (2012) and the methodology described in Nakamura et al. (2013).

TABLE 4—QUANTILE REGRESSION EFFECTS OF COUNTRY-SPECIFIC GEOPOLITICAL RISK

	GDP growth (t+1) (1)	TFP growth (t+1) (2)	Military exp. (t+1) (3)
OLS	−0.35 (0.22)	−0.22 (0.27)	2.15 (0.39)
Quantile			
q50	−0.24 (0.22)	−0.04 (0.14)	0.63 (0.19)
q10	−1.44 (0.63)	−1.86 (0.45)	0.16 (0.03)
q90	0.30 (0.30)	1.53 (0.55)	7.08 (0.55)
Observations	3,082	2,261	2,681
Countries	26	19	26

Notes: Standard errors, in parentheses, are bootstrapped using 500 replications. The table shows quantile regression effects of geopolitical risk in a panel of countries from 1900 through 2019. In each specification, the right-hand side variable in country-specific GPR in year t (standardized by country). The dependent variables are GDP growth, TFP growth, and military expenditures in year $t+1$, respectively. GDP growth and TFP growth are expressed in percent units. Military expenditures are expressed as a share of GDP. The OLS coefficients are reported in the top row. The quantile coefficients report the effects at the fiftieth, tenth, and ninetieth percentile of the distribution of the dependent variable. All regressions include an intercept and control for global geopolitical risk. Real GDP per capita data are from Barro and Ursua (2012), extended through 2019 using the World Bank World Development Indicators. TFP data are from Bergeaud, Cetté, and Lecat (2016). Military expenditures are taken from Roser and Nagdy (2013).

Economic message: The real effects of GPR show up in firm investment decisions, consistent with precautionary delay or worsening expected profitability.

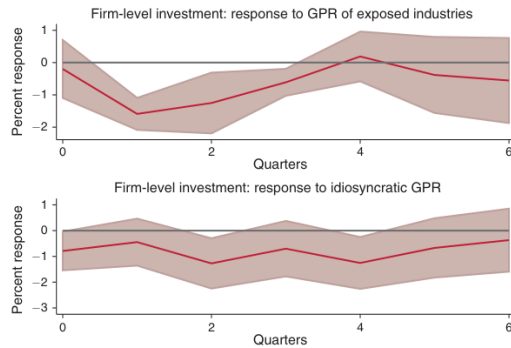


FIGURE 11. RESPONSE OF FIRM-LEVEL INVESTMENT TO GEOPOLITICAL RISK

The top panel plots the dynamic response of investment following a two standard deviation increase in GPR for a firm in an industry with positive exposure to geopolitical risk. The bottom panel plots the response of investment following a two standard deviation increase in firm-level GPR. The shaded areas 90 percent confidence intervals. Standard errors are two-way clustered by firm and quarter-industry.

TABLE 5—GEOPOLITICAL RISK AND FIRM-LEVEL INVESTMENT

$IK(t+2)$	(1)	(2)	(3)	(4)
$\Delta GPR \times$ dummy industry exposure	−0.63 (0.29)	−0.64 (0.27)		
GPR firm level			−0.67 (0.30)	
ΔGPR	−1.39 (1.19)			
Political risk Hassan et al. (2019)				−0.75 (0.25)
Cash flow	2.72 (0.46)	2.78 (0.46)	2.67 (0.38)	2.48 (0.30)
Tobin's Q	8.91 (1.68)	7.93 (1.56)	9.31 (0.92)	9.47 (0.90)
$IK(t-1)$	0.31 (0.01)	0.30 (0.01)	0.24 (0.01)	0.26 (0.01)
Observations	374,727	374,727	95,073	112,161
Firm fixed effects	Yes	Yes	Yes	Yes
Time effects	No	Yes	Yes	Yes
R^2	0.45	0.47	0.59	0.58
Sample	85Q1–19Q4	85Q1–19Q4	05Q1–19Q4	05Q1–19Q4

Notes: Standard errors, in parentheses, are clustered by industry and quarter in columns 1 and 2, by firm and quarter-industry in columns 3 and 4. The table shows results from regressions of firm-level investment on geopolitical risk at the industry or at the firm level. The dependent variable IK is defined as $100 \times \log ik$, where ik is the ratio of capital expenditures to previous-period property, plant, and equipment, as defined in the text. All variables (except the dummy exposure variable) are standardized.

5.10 Table 5: Geopolitical risk and firm-level investment

Read: The interaction of aggregate GPR with industry exposure has a negative coefficient of about -0.63 to -0.64 . Firm-level GPR has a negative coefficient of about -0.67 . Firm-level political risk from Hassan et al. has a similar coefficient of about -0.75 .

Economic message: The firm-level results are economically meaningful and comparable in magnitude to the effect of broader political risk.

6 Short summary

- The paper constructs a transparent news-based geopolitical risk index covering 1900 onward.
- The index measures the share of newspaper articles discussing adverse geopolitical events, and it separates threats from acts.
- The index lines up well with major historical episodes, a manually coded narrative index, war deaths, military spending news, and country-specific events.
- In US VARs, a GPR shock predicts lower investment, lower hours, lower stock prices, and lower oil prices.
- Both realized acts and threats reduce activity, implying that fear and anticipation are economically important.
- In a long cross-country panel, higher GPR is associated with higher disaster probability and greater downside risk to GDP growth.
- In firm data, GPR reduces investment more for exposed industries and for firms whose earnings calls mention idiosyncratic geopolitical risk.

7 Conclusion

7.1 Core conclusion

Caldara and Iacoviello show that geopolitical risk can be measured systematically using newspaper text, and that the resulting measure has meaningful macroeconomic and firm-level implications. Higher GPR foreshadows lower investment and employment, higher disaster probability, larger downside risks to GDP growth, and lower firm-level investment.

7.2 Economic intuition

Geopolitical risk affects the economy through several channels:

- **Uncertainty and precaution:** firms delay investment and hiring when future geopolitical outcomes are uncertain.
- **Expected demand and profitability:** conflicts and tensions can reduce expected sales, disrupt supply chains, and raise costs.
- **Financial-market repricing:** asset prices quickly reflect industry exposure to geopolitical shocks.
- **Tail-risk reassessment:** geopolitical tensions make severe economic outcomes more likely.
- **Realized damage:** wars and terrorist events can destroy capital, interrupt trade, and redirect resources toward military activity.

7.3 Takeaway

The paper's main lesson is that geopolitical risk is measurable and economically consequential. The threat of adverse events is itself a real economic force, and realized acts reinforce that force. The GPR index gives researchers a tool for studying those channels over more than a century and at aggregate, country, industry, and firm levels.